

# Common Knee Injuries and Conditions

## Patellofemoral Pain Syndrome

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| <b>Injury overview</b>                                         | This is a general term used to describe a broad range of pain in the anterior aspect of the knee.                                                                                                                      |
| <b>Surgical and rehabilitative management</b>                  | Surgical intervention is typically not required. Initial rehabilitation will focus on inflammation management and gentle exercises through pain-free ranges of motion.                                                 |
| <b>Return-to-play exercise and conditioning considerations</b> | Exercises should focus on correcting altered biomechanical factors contributing to poor movement quality of the hip, knee, and foot/ankle, and improving the efficiency of multi-joint, multiplanar movement.          |
| <b>Return-to-play criteria</b>                                 | Hip extension ROM (0 degrees) when combined with knee flexion, hip internal rotation ROM (45 degrees), ankle dorsiflexion ROM (20 degrees). Completion of sport-specific functional test battery without complication. |
| <b>Estimated timeline</b>                                      | Most cases will resolve to full participation within 1–4 weeks.                                                                                                                                                        |

## Patellar Tendinopathy

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| <b>Injury overview</b>                                         | Common in sports with an emphasis on jumping cutting, such as basketball, soccer, and volleyball. Patellar tendonitis (“jumper’s knee”) is an overuse injury of the patellar tendon, which produces pain at the anterior aspect of the patella and often localized tenderness to the inferior aspect of the patella. |
| <b>Surgical and rehabilitative management</b>                  | Immobilization may be recommended when a painful ROM exists. Crutches may alleviate pain during weight-bearing. Rehabilitation begins quickly within pain-free ROMs to control inflammation and restore normal muscle activation patterns. Surgical interventions are rare.                                          |
| <b>Return-to-play exercise and conditioning considerations</b> | Flexibility to address functional limitations and restore normal range of knee extension (with the hip flexed), ankle dorsiflexion (straight and bent knee), and hip internal rotation. A slow introduction of eccentric training for the quadriceps.                                                                |
| <b>Return-to-play criteria</b>                                 | Pain-free ROM and comparable strength symmetry. Hip internal ROM (45 degrees); hip extension 90/90 (20 degrees); ankle dorsiflexion ROM (20 degrees). Sport-specific functional test battery completed without complications.                                                                                        |
| <b>Estimated timeline</b>                                      | Conservatively, symptoms usually subside within 1–4 weeks.                                                                                                                                                                                                                                                           |

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## Osteoarthritis (OA)

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| <b>Injury overview</b>                                         | This chronic condition involves degeneration of the articular cartilage, which will reduce the intra-articular space, affect joint stability and surface integrity, cause pain, and affect the long-term quality of life and disability. OA results from chronic overuse, altered joint mechanics, and poor movement quality.                  |
| <b>Surgical and rehabilitative management</b>                  | Surgical interventions do not eliminate OA but may effectively mitigate current mechanisms contributing to symptoms (i.e., removal of bone spurs, reshaping of bony structures, microfracture procedure to produce postsurgical scarring effect to “regenerate” the smooth articular surface).                                                 |
| <b>Return-to-play exercise and conditioning considerations</b> | Pain-free ROM and comparable strength symmetry. Hip internal ROM (45 degrees); hip extension 90/90 (25 degrees); ankle dorsiflexion ROM (20 degrees). Sport-specific functional test battery completed without complications (including the star balance excursion test, shark skill test, triple hop for distance, 5–10–5 test, and T-drill). |
| <b>Return-to-play criteria</b>                                 | Continue progressing overall metabolic specificity and conditioning to preserve core strength, balance, upper-body strength and power, speed, and agility.                                                                                                                                                                                     |
| <b>Estimated timeline</b>                                      | Return to participation is usually a self-limited scenario. Typically, athletes can return once symptoms are managed.                                                                                                                                                                                                                          |

## Anterior Cruciate Ligament (ACL) Sprain

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| <b>Injury overview</b>                        | Sprain of this ligament is common with rapid deceleration or change of direction with knee valgus position and planted foot, typically without contact.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Surgical and rehabilitative management</b> | <p>Following an acute injury, there is a period of non-weight-bearing until pain-free ROM and normal muscle activation of the quadriceps are regained. Some injuries can be managed through comprehensive exercise training, sports participation limitations, and protective bracing. Surgical interventions are common and, when performed early, restore dynamic stability of the joint and minimize the risks of OA development (long term).</p> <p>Rehabilitation focuses on inflammation and pain management, progressing to therapeutic exercises to restore normal ROM and joint arthrokinematics. Closed- and open-chain exercises are then introduced to improve the ability to produce and reduce forces dynamically.</p> |

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| <b>Return-to-play exercise and conditioning considerations</b> | Continue progressing overall metabolic specificity and conditioning to preserve core strength, balance, upper-body strength and power, speed, and agility.                           |
| <b>Return-to-play criteria</b>                                 | Sport-specific functional performance battery without complication (including the star balance excursion test, shark skill test, triple hop for distance, 5–10–5 test, and T-drill). |
| <b>Estimated timeline</b>                                      | Nonsurgical: 2–4 weeks. Surgical: 6–9 months.                                                                                                                                        |

## Medial Collateral Ligament (MCL)

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| <b>Injury overview</b>                                         | Injury to the supporting ligament to the inner (medial) side of the knee. Typically, these occur from a direct blow to the outside of the knee, creating pain to the medial joint surface and producing feelings of joint instability.                                                                                                                                                                                                                          |
| <b>Surgical and rehabilitative management</b>                  | Surgical interventions are generally not required if the injury is contained to the MCL. Should the damaged tissues include the ACL and meniscus, it is a likely case for surgical repair. Rehabilitation programs will include an initial focus on inflammation and pain management, restoration of flexibility and balance deficits, and inclusion of progressive resistance exercises. Using a protective (hinged) knee brace is considered a best practice. |
| <b>Return-to-play exercise and conditioning considerations</b> | Continue progressing overall metabolic specificity and conditioning to preserve core strength, balance, upper-body strength and power, speed, and agility.                                                                                                                                                                                                                                                                                                      |
| <b>Return-to-play criteria</b>                                 | Sport-specific functional performance battery without complication (including the star balance excursion test, shark skill test, triple hop for distance, 5–10–5 test, and T- drill).                                                                                                                                                                                                                                                                           |
| <b>Estimated timeline</b>                                      | 1–2 weeks for minor sprain, extending to 3 months for more severe sprains (with or without surgical intervention). If surgical repair includes other structures, this timeline is extended.                                                                                                                                                                                                                                                                     |

## Posterior–Lateral Complex

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| <b>Injury overview</b>                        | This term encompasses injuries to the lateral collateral ligament (LCL), posterior cruciate ligament (PCL), lateral meniscus, and proximal fibular head fractures. Injuries that involve only the LCL are rare; more structures are usually damaged.   |
| <b>Surgical and rehabilitative management</b> | Surgical interventions are common to rapidly restore joint stability and normal movement biomechanics. Nonsurgical treatments are also an option, though the lack of joint stability increases OA risks (which impacts the long-term quality of life). |

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| <b>Return-to-play exercise and conditioning considerations</b> | Continue progressing overall metabolic specificity and conditioning to preserve core strength, balance, upper-body strength and power, speed, and agility.                                                                                               |
| <b>Return-to-play criteria</b>                                 | Sport-specific functional performance battery without complication (including the star balance excursion test, shark skill test, triple hop for distance, 5–10–5 test, and T-drill).                                                                     |
| <b>Estimated timeline</b>                                      | 1–2 weeks for a minor sprain, extending up to a year depending on the specific structures damaged and the time between the injury date and the surgery date. The best outcomes are achieved when the repair is within 2 weeks of the initial injury date |

## Meniscal Tear

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| <b>Injury overview</b>                                         | Under normal conditions, the medial and lateral meniscus provides joint cushioning and stability and allows for smooth joint movement. The vascular structure provides some complications, as not all sections of the meniscus have good blood flow. The location of the lesion will determine whether the injury can heal without surgical intervention.                                                                     |
| <b>Surgical and rehabilitative management</b>                  | Surgical repairs include suturing or removing meniscal sections to create a smooth joint surface. Rehabilitative management is initially dependent on the specific surgical procedures performed. Partial meniscectomy generally has the athlete back to controlled weight-bearing activities within 1 week. In contrast, those with meniscal repairs require straight-leg immobilization and non-weight-bearing for 4 weeks. |
| <b>Return-to-play exercise and conditioning considerations</b> | Continue progressing overall metabolic specificity and conditioning to preserve core strength, balance, upper-body strength and power, speed, and agility.                                                                                                                                                                                                                                                                    |
| <b>Return-to-play criteria</b>                                 | Sport-specific functional performance battery without complication (including the star balance excursion test, shark skill test, triple hop for distance, 5–10–5 test, and T-drill).                                                                                                                                                                                                                                          |
| <b>Estimated timeline</b>                                      | Conservative approach: 2–6 weeks. Surgical partial meniscectomy: 4–6 weeks. Surgical meniscal repair: 10 weeks, up to 6 months.                                                                                                                                                                                                                                                                                               |